Machine Learning for Maternal Risk Monitoring

61.501 Production-ready ML | Project Proposal | 15 October 2026

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1. Research Question and Significance

Maternal health risk monitoring is essential for identifying women who may need closer assessment and timely care during pregnancy. However, risk can arise from the combined effects of age, blood pressure, blood glucose and other physiological indicators, making it difficult to assess through isolated thresholds or manual review alone. Machine learning offers a way to learn relationships among these indicators and integrate them into a consistent risk classification. This project compares classical machine learning models for maternal risk classification, exploring their potential to support screening and prioritize limited healthcare resources in underserved communities.

2. Dataset Preparation

We will use **UCI Maternal Health Risk** [1], collected in Bangladesh, with approximately 1,000 labeled records. Inputs are age, systolic and diastolic blood pressure, blood glucose, body temperature and heart rate; the output is **low risk, mid risk or high risk**. We will record the data version, inspect units and implausible values, remove exact duplicate rows and audit conflicting labels. No new data collection is required.

3. Experiments and Evaluation Criteria

An example data record is shown below.

Age (years)	Systolic BP (mmHg)	Diastolic BP (mmHg)	Glucose (mmol/L)	Temperature (°F)	Heart rate (bpm)	Risk label
25	130	80	15	98	86	high risk

Models: Logistic Regression, RBF Support Vector Machine, Gaussian Naive Bayes, Decision Tree, Random Forest and Gradient Boosting (the latter two are tree ensembles). These six models use course topics; a majority-class predictor provides the baseline.

Protocol: Reserve approximately 20% for final testing. Tune hyperparameters on the remainder using five-fold stratified group cross-validation, subject to class/group counts. Keep identical input vectors in the same split and fold; fit preprocessing within folds, including scaling for Logistic Regression and SVM. Select the model by validation **macro-F1**, then evaluate it once on the held-out test set.

Measures: Report accuracy, macro-F1, per-class precision/recall, the confusion matrix, cross-validation variability and inference latency. Emphasize high-risk recall, inspect misclassifications, and compare errors across age groups with group sizes reported. Patient identifiers are unavailable, limiting independence claims.

4. Expected Outcomes

- **Reproducible code for maternal risk classification models.** Following data preprocessing, the project will train, tune and evaluate several maternal risk classification models.
- **Model comparison and analysis report.** The project will present each model's performance, explain the choice of the final model, and analyze error cases and limitations.
- **Simple prediction demo and presentation materials.** The project will use the most suitable final model to build a demo that accepts a batch of data records and displays their predicted risk categories, accompanied by the final report and presentation slides.

5. Schedule

Period	Planned work
15-20 Oct	Confirm scope, audit data and submit proposal by 20 Oct, 23:59.
21 Oct-1 Nov	Preprocess data, create group-aware splits and implement reproducible model training code.
2-15 Nov	Tune and evaluate models; select the final model and draft the comparison and error analysis report.
16-29 Nov	Build and test a batch prediction demo using the selected model; discuss limitations and verify reproducibility.
30 Nov-6 Dec	Finalize code, report and slides; submit by 6 Dec, 23:59. Presentation: Week 13 (TBC).

[1] Ahmed, M. (2020). *Maternal Health Risk*. UCI Machine Learning Repository. doi:10.24432/C5DP5D. CC BY 4.0.